

```
$ curl /users/123 | usecase | repository | db
```

Clean Architecture

// 0000000000

Controller

□□

Usecase

□□

Repository

□□

Presenter

□□□

\$ cat blueprint.md



01 /



02 /



03 /



TypeScript



04 /



DB



PART 01



□□□□□□□□



```
1 // UI 로직
2 app.get('/users', (req, res) => {
3   const users = db.query('SELECT * FROM users');
4   res.json(users.map(u => ({
5     name: u.name,
6     email: u.email,
7     createdAt: u.created_at.format('YYYY-MM-DD')
8   })));
9 });
```

INCOMING REQUESTS

GraphQL

→

→ 10

MongoDB

→ SQL

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□□□□□



Repository



API □□□



Presenter

□□□□□□



Controller

INVARIANT

□□□□Usecase□□□□ — □□□□□□□□□□□□□□□□

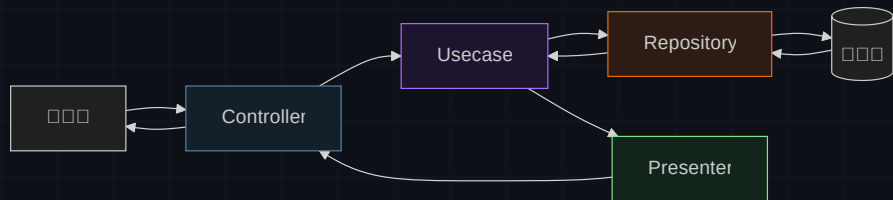
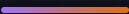
PART 02



MySQL MongoDB API

[illegible]

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□□

UI → Usecase → Repository → DB

□□

DB → Repository → Usecase → Presenter → UI □□□□

PART 03

TypeScript □□

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Repository

```
1 export interface UserRepository {
2   getUser(id: number): Promise<User | null>;
3 }
4
5 // MySQL — Usecase
6 export class MySQLUserRepository
7   implements UserRepository {
8   async getUser(id: number) {
9     const rows = await db.query(
10       'SELECT * FROM users WHERE id = ?', [id]);
11     return rows[0] ?? null;
12   }
13 }
```

Usecase

```
1 export class GetUserUseCase {
2   constructor(
3     private userRepository: UserRepository
4   ) {}
5
6   async execute(id: number) {
7     const user =
8       await this.userRepository.getUser(id);
9     if (!user) return null;
10
11     // 30 days ago
12     if (user.createdAt < thirtyDaysAgo()) {
13       return null;
14     }
15     return user;
16   }
17 }
```

    UserRepository — MySQL   Mongo    

Presenter NO GRAPH Controller

```
1 // Presenter
2 export class UserPresenter {
3   static toResponse(user: User | null) {
4     if (!user) return { error: 'User not found' };
5     return {
6       id: user.id,
7       name: user.name,
8       created_at: user.createdAt.toISOString(),
9     };
10  }
11 }
```



```
1 // routes/user.ts — 
2 const repo = new MySQLUserRepository();
3 const usecase = new GetUserUseCase(repo);
4 const controller = new UserController(usecase);
5 
6 router.get('/users/:id', controller.getUser);
```

COMPOSITION ROOT

Controller NO GRAPH Usecase NO GRAPH — NO GRAPH

PART 04



MongoDBUserRepository new

```
Presenter toResponse()
```

CacheUserRepository NO GROUPS NO GROUPS MySQLUserRepository NO GROUPS NO GROUPS NO GROUPS NO GROUPS NO GROUPS NO GROUPS NO GROUPS NO GROUPS NO GROUPS NO GROUPS NO GROUPS DBUsecase NO GROUPS NO GROUPS NO GROUPS NO GROUPS

\$ 000000000 — 000000000000

CHECKPOINT_01

다음의 코드에서 Cache를 사용하는 부분은?

A Usecase에서 Cache를 사용한다

B Controller에서 Cache를 사용한다

C Repository에서 CacheUserRepository를 사용하여 DB

FAQ

MVC

MVC Clean Architecture

10 100 1 100

2

Repository

Usecase



Layers separated.

□□□□Spring Boot × DDD□□□□□□□□□□□□

```
$ curl /users/123
```

```
controller → usecase → repository → presenter
```